

Supplementary Material

Learning the Lab, Not the Law: An Audit Framework for Structure-Level Domain Confounding in Symbolic Scientific Discovery

This supplement contains material referenced from, but not essential to following the argument of, the main text: extended derivations, the full per-source data-provenance detail, a secondary ablation figure, and additional analyses (Sections S4–S7). Sections S8–S13 were added in the first revision: the extended Stage 0 study, the complete verdict case analysis, detail on the S_5 selection, the specificity experiments of the submitted version, notes on the corrected conformal evaluation, and the full tables of the simulation benchmark. Every number here is backed by a script or an archived output in the released repository; `run_all.sh` specifically regenerates the S4 and S6 analyses (Sections S4 and S6 below), and none is estimated or asserted without a corresponding script or archived output.

1 S1: Few-cluster inference, extended

Section 3.5 of the main text (Component 3: Hierarchical cluster bootstrap) states that with $F = 4$ facilities the naive percentile cluster bootstrap is not, on its own, a frequentist-valid interval, and that the Ibragimov–Müller estimator is used as the primary inferential basis instead. The full justification: Cameron et al. [2] validate the wild cluster bootstrap from five clusters upward; MacKinnon and Webb [5] show it can fail with very few or badly unbalanced clusters; Canay et al. [3] prove a fixed-number-of-clusters over-rejection bound that, at $q = 4$ clusters, reaches $2^{1-q} = 0.125$ — up to 12.5 percentage points of over-rejection above the nominal level. This is why no conclusion in the main text rests on the cluster-bootstrap interval alone.

As a further check, we ran a Monte-Carlo calibration of the cluster-bootstrap’s own coverage under a known data-generating process matched to the real cluster sizes ($n_f = 45, 41, 20, 8$): empirical coverage was 0.994 against a 0.95 nominal target. The cluster bootstrap is therefore *over*-conservative here (valid but unnecessarily wide) rather than anti-conservative, so it is kept in the main text as an honest, if loose, descriptive comparison among resampling schemes, while the Ibragimov–Müller interval carries every claim about the exponent’s sign.

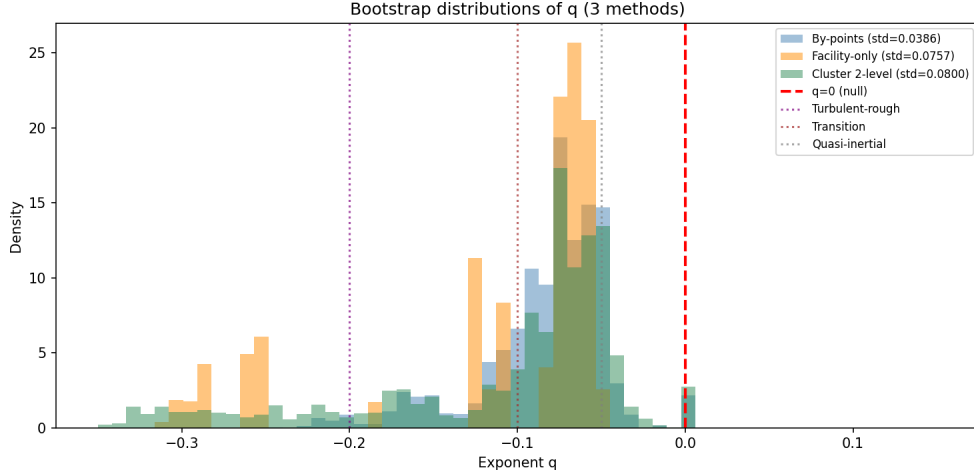


Figure 1: Bootstrap distributions of the scaling exponent q under three resampling schemes, overlaid on a single panel (Table 6). The by-points scheme yields a deceptively tight interval; facility-only and two-level cluster resampling, which correctly account for the four-facility structure, are both far wider — the gap is a direct measurement of apparent precision lost to facility clustering.

2 S2: Data provenance, full detail

Table 1 reproduces the main text’s Table “provenance” (Appendix A, Reproducibility) with the complete per-source extraction narrative; the main text keeps a condensed version.

3 S3: Adaptive-conformal ablation, extended

Section 6.1 of the main text (Discussion: ablations) summarizes the adaptive-conformal ablation in two sentences. Fig. 2 is the full figure: learned clusters (via unsupervised clustering on the covariates) simply rediscover facility identity, and the resulting prediction intervals are $\sim 25\%$ wider than normalized conformal’s with a LOGO-minimum of only 0.439 — *worse* than the facility-agnostic normalized-conformal baseline the main text recommends. This is reported as a negative result: letting the model additionally choose its own grouping does not help, and does not need to be re-derived to follow the main argument.

4 S4: Per-facility error in original units

The main text’s Table “baselines” reports $R_{\log}^2$ and R_{orig}^2 (aggregate, in-sample). Table 2 breaks the same in-sample fits down by facility, in original C_p units (MAPE and RMSE), computed from the identical fitted values underlying R_{orig}^2 — not a re-fit, not LOGO-CV (that is already a separate column in the main-text table).

Two things stand out that the aggregate columns hide. First, Class-SR’s error is markedly uneven across facilities in original units (MAPE from 4.8% on Zheng2024 to 22.5% on Vrancik1968, a $4.7\times$ spread), consistent with the cross-facility heterogeneity the paper’s central claim describes rather than a uniformly good global fit. Second, C_p constant per facility (a model that literally memorizes each facility’s mean) is competitive with the global power law on Liu2024 and Zheng2024 but catastrophically worse on Xia2024 (61.9% MAPE) — a reminder that no single baseline is

Table 1: Full data provenance by source (extended version of the main-text table).

Source	Reference (DOI)	n	Extraction detail	Redistribution
Vrancik1968	NASA TN D-4849 (1968)	41	All 41 points digitized from Fig. 13 (torque-coefficient vs. Reynolds-number curves for 4 rotor configurations); reading error estimated at 5–15% (mean 8.7%) from digitization-tool pixel resolution against the plotted axis range	Public domain (US Government report)
Xia2024	<i>Appl. Sci.</i> 14(17):7613; 10.3390/app14177613	45	37/45 points digitized from figures (windage-power-vs-speed curves across 5 rotor geometries); 8/45 taken from the paper’s reported CFD-validated closed-form fit rather than a plotted curve; err. 3–28.7% (mean 6.8%), highest for the lowest-speed points where absolute power is small and relative digitization error is amplified	CC-BY 4.0
Liu2024	Research Square preprint; 10.21203/rs.3.rs-4237491/v1	20	15/20 digitized from Fig. 7; 5/20 computed from the constant given in the paper’s Eq. 12; err. 5–10% (mean 6.3%)	CC-BY 4.0 (Research Square’s standard preprint license)
Zheng2024	<i>J. Zhejiang Univ.-SCI A</i> 25(11):938–952; 10.1631/jzus.A2300288	8	All 8 points are direct transcription of measured values from the source’s Table 1 (no digitization); the 1.4–83.1% range reported for this source is the <i>source authors’</i> own theoretical-model-vs-measurement discrepancy, not an uncertainty of our transcription, and is largest at the lowest rotational speed where windage power is smallest and relative discrepancy is correspondingly amplified	© publisher; derived numerical values only

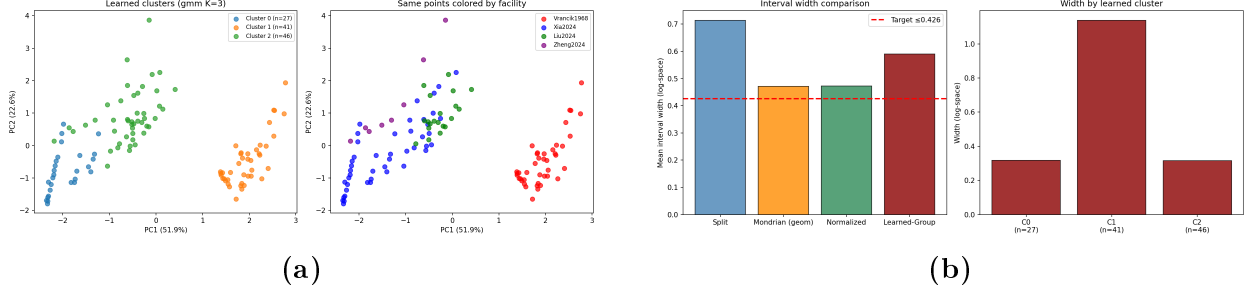


Figure 2: Adaptive-conformal ablation. **(a)** Clusters learned from the data (colour) overlaid on facility identity: the learned grouping rediscovers the facilities rather than a physical structure — the adaptation buys nothing. **(b)** Interval widths for the learned-group (adaptive) conformal variant versus the baselines: adaptation *widens* intervals ($\sim 25\%$) while failing the LOGO-minimum acceptance criterion (main-text Table “conformal”).

Table 2: Per-facility MAPE (%) and RMSE (original C_p units) for each baseline in Table “baselines”, computed from the same in-sample fitted values as R_{orig}^2 . `scripts/facility_mape_rmse.py` in the released repository regenerates this table.

Model	Liu2024	Vrancik1968	Xia2024	Zheng2024	All
<i>MAPE (%)</i>					
Class-SR	6.9	22.5	5.9	4.8	11.9
C_p global power	44.2	33.7	82.1	33.2	54.6
C_p const. facility	7.6	51.0	61.9	6.7	44.6
C_p const. geometry	7.6	27.2	6.6	6.7	14.2
<i>RMSE (original C_p units)</i>					
Class-SR	0.0067	0.0426	0.0015	0.0050	0.0257
C_p global power	0.0406	0.0442	0.0086	0.0292	0.0329
C_p const. facility	0.0072	0.0703	0.0117	0.0070	0.0429
C_p const. geometry	0.0072	0.0474	0.0019	0.0070	0.0287

uniformly informative across facilities, which is itself evidence against treating any one of them as “the” physical model.

5 S5: Naive/blind symbolic-regression replication of the S5 finding

A natural question about Section 5.7 of the main text (the S_5 case study) is whether an unbiased, off-the-shelf symbolic regression search independently rediscovers the same Mach-threshold structure, or whether finding it required deliberately including a regime-switching candidate in the hand-designed S_1 – S_6 family. We ran `gplearn`’s genetic-programming symbolic regressor (population 3,000, 40 generations, arithmetic primitive set only, no piecewise or threshold primitive available to it) on the same residual target the S_1 – S_6 family operates on ($\log C_p - \log C_i$, i.e. after removing the per-geometry prefactor already fit by class-SR), with $\log(\text{Re}_\Omega)$ and Mach as its only inputs and no access to facility or geometry labels.

Result: it does not. The fittest program after 40 generations is $-0.075 \times \log(\text{Re}_\Omega)$ (in-sample $R^2 = 0.470$) — a plain single-exponent power law that never incorporates Mach at all. An unconstrained genetic search, with no regime-switching primitive handed to it and no hint that a threshold structure might be worth trying, converges to the simplest fit and does not stumble onto the two-regime artifact.

We read this as a genuine scope clarification, not a weakness the main text should downplay: the S_5 vulnerability specifically requires a structure-selection search whose *candidate library* already includes regime-switching forms (as ours does, and as increasingly capable automated hypothesis-generation pipelines — the kind this protocol is meant to audit — routinely do), not every symbolic-regression pipeline regardless of search-space design. A minimal, generic evolutionary search with a narrow primitive set will not independently reproduce this failure mode; a Bayesian search over an expanded candidate family that includes threshold/regime structures can. `scripts/naive_sr_replication.py` in the released repository reproduces this result exactly.

6 S6: Synthetic power curve, structural (not prefactor) confounding

Caveat (first revision). This experiment uses the simplified classifier described at the start of Section S11, not the full protocol; it is superseded by regime R4a of the simulation benchmark (Section S13).

The main text’s specificity/power analysis (Section S11, Fig. 4) injects a *prefactor*-level confound (a constant additive offset on one facility’s $\log C_p$) and reports low detection power by design, because class-SR’s per-geometry prefactor absorbs it. Here we inject a *structural* confound instead: one facility’s Reynolds exponent itself is shifted by Δq relative to the other three, and — motivated by the hypothesis that S_5 might reflect one facility (Liu2024) occupying its own Mach range, a mechanism that the main text argues does not explain S_5 , since removing Liu2024 does not remove the structure — that facility’s Mach values are also drawn from a disjoint range, so the exponent shift is correlated with Mach exactly as the S_1 – S_6 Bayesian search could pick up on. (An earlier version of this experiment gave the confounded facility the same random Mach range as everyone else; since the S_1 – S_6 family only searches for regimes defined *on Mach*, an exponent shift uncorrelated with Mach gave it nothing to find, and measured power sat at the false-positive floor for every Δq — a bug in the experiment, not a finding about the protocol, corrected before the results below.)

Two things are worth noting honestly. First, power at $\Delta q = 0$ (no real exponent difference at all) is already 0.13, not 0: the confounded facility’s disjoint Mach range alone — with no genuine

Table 3: Detection power (fraction of $K = 100$ synthetic datasets flagged facility-artifact / statistically-real-spurious / report- with-disclaimer) as the injected exponent gap Δq grows, with the confounded facility’s Mach range disjoint from the other three. `scripts/structural_confound_power.py` in the released repository regenerates this table.

Δq	Power	95% CI (Clopper–Pearson)	Dominant verdict
0.00	0.13	[0.07, 0.21]	no-regime-detected (87/100)
0.02	0.17	[0.10, 0.26]	no-regime-detected (83/100)
0.05	0.16	[0.09, 0.25]	no-regime-detected (84/100)
0.10	0.30	[0.21, 0.40]	no-regime-detected (70/100)
0.20	0.70	[0.60, 0.79]	ambiguous (70/100)

structural difference – is enough to occasionally trigger an “ambiguous” or “facility-artifact” verdict, since Test 1 (threshold-vs-facility contingency) reacts to the covariate separation itself. This is consistent with Test 1 being, on its own, a qualitative flag rather than a stand-alone significance gate (Section 5.7 of the main text). Second, the dominant verdict at the largest gap tested ($\Delta q = 0.2$) is *ambiguous* rather than *facility-artifact* outright: the protocol increasingly refuses to call a clean verdict as the confound grows, rather than confidently misclassifying it as physics, which is the behaviour the protocol is meant to have in a genuinely uncertain regime.

7 S7: Pairwise covariate-overlap (MMD)

Section 3.1 of the main text (Problem formalization) quantifies the extreme covariate shift across facilities with the scalar characteristic-scale ratio $\Lambda = 70.4$. Table 4 complements this with a direct, multivariate quantification: the unbiased, RBF-kernel maximum mean discrepancy (median-heuristic bandwidth) between every pair of facilities on the joint $(\log \text{Re}_\Omega, \text{Ma})$ covariate space, each standardized to zero mean/unit variance before computing the kernel.

Table 4: Pairwise squared MMD on standardized $(\log \text{Re}_\Omega, \text{Ma})$. Larger values indicate less covariate overlap. Facility sample sizes in parentheses.

Facility pair	n_1, n_2	MMD ²
Liu2024 vs. Vrancik1968	20, 41	0.101
Liu2024 vs. Zheng2024	20, 8	0.125
Vrancik1968 vs. Zheng2024	41, 8	0.158
Xia2024 vs. Zheng2024	45, 8	0.634
Liu2024 vs. Xia2024	20, 45	0.757
Vrancik1968 vs. Xia2024	41, 45	0.913

All six pairs show non-trivial separation; none is close to 0 (complete overlap). The ordering does not simply track the univariate Mach-threshold story of Section 5.7 (Liu2024 is not the most separated facility in this joint space; Vrancik1968 and Xia2024 are) — a reminder that the S_5 discriminant battery targets one specific, Mach-conditioned mechanism, not overall covariate distance, and that a facility pair can be jointly far apart in (Re, Ma) space without that separation aligning with any structure a candidate family searches for.

8 S8: Stage 0 synthetic validation, extended

This section gives (a) the by-factor results and threshold sensitivity of the phase study summarised in Section 5.2 of the main text (script `revision/stage0_phase.py`, seed 20261002, 300 designs per cell, 14,400 in total), and (b) the four earlier experiments of the submitted version (script `analysis/stage0_validation.py`), retained for completeness.

Generator of the phase study. Each facility design is a full factorial over 3–8 rotational speeds (linearly spaced in $[40, 100]$ rad/s and scaled by a random factor in $[0.7, 1.3]$), two radii (relative span $10^{U(-3.5, 0.5)}$, scaled by a random factor in $[0.5, 2]$), two pressures (span $10^{U(-3.5, 1)}$) and, when four controls are varied, two temperatures (span $10^{U(-3.5, -0.3)}$). Imbalance duplicates one design point as many times as there are runs. The second facility, when present, has its own random spans and an additive response shift of 0 or 0.3. True effects: $\theta_{\text{Re}} \sim U(-0.6, 0)$, $\theta_{\text{Ma}} \sim U(-0.3, -0.05)$, $\theta_{N_g} \sim U(-0.15, 0)$; noise $\sigma = 10^{U(-2.5, -1)}$. The response is fitted by least squares with facility fixed effects.

Table 5: Stage 0 phase study by factor (frozen rules: $\kappa \leq 30$, pairwise $|r| < 0.99$, recovery tolerance $c = 0.5$). Blind zone: all pairwise $|r| < 0.99$ but $\kappa > 30$. False promise: share of accepted designs that are not recoverable.

Factor	Level	n	Recov.	Blind zone	Recov. in blind	Acc. S0	Acc. pairwise	False prom. S0 / pw
speeds	3	3600	0.334	0.411	0.240	0.795	0.579	0.246 / 0.569
speeds	4	3600	0.352	0.411	0.243	0.799	0.576	0.202 / 0.557
speeds	6	3600	0.381	0.417	0.263	0.785	0.574	0.186 / 0.534
speeds	8	3600	0.409	0.417	0.297	0.774	0.594	0.155 / 0.498
controls varied	3	7200	0.310	0.412	0.209	0.817	0.570	0.226 / 0.597
controls varied	4	7200	0.428	0.417	0.313	0.759	0.592	0.173 / 0.485
replication imbalance	0	7200	0.357	0.409	0.230	0.804	0.578	0.215 / 0.548
replication imbalance	1	7200	0.381	0.420	0.291	0.773	0.584	0.175 / 0.529
facilities	1	4800	0.221	0.209	0.182	0.849	0.706	0.247 / 0.616
facilities	2	9600	0.443	0.517	0.277	0.758	0.518	0.186 / 0.524
facility shift	0.0	9600	0.329	0.366	0.251	0.803	0.612	0.207 / 0.552
facility shift	0.3	4800	0.449	0.510	0.276	0.759	0.519	0.182 / 0.520

Table 6: Accuracy (and false-promise rate) of each screen as a function of its threshold and of the recovery tolerance c .

Rule	$c = 0.2$	$c = 0.3$	$c = 0.5$	$c = 0.7$
Stage 0, $\kappa \leq 10$	0.778 (0.275)	0.729 (0.211)	0.660 (0.102)	0.615 (0.068)
Stage 0, $\kappa \leq 20$	0.818 (0.343)	0.796 (0.256)	0.757 (0.152)	0.726 (0.104)
Stage 0, $\kappa \leq 30$	0.816 (0.394)	0.810 (0.302)	0.788 (0.197)	0.770 (0.139)
Stage 0, $\kappa \leq 50$	0.792 (0.455)	0.805 (0.361)	0.808 (0.246)	0.803 (0.184)
Stage 0, $\kappa \leq 100$	0.729 (0.535)	0.759 (0.445)	0.791 (0.327)	0.804 (0.261)
pairwise, $ r < 0.95$	0.714 (0.660)	0.683 (0.583)	0.634 (0.492)	0.602 (0.435)
pairwise, $ r < 0.99$	0.513 (0.692)	0.543 (0.628)	0.581 (0.539)	0.603 (0.484)
pairwise, $ r < 0.999$	0.410 (0.715)	0.456 (0.654)	0.521 (0.567)	0.562 (0.513)

Earlier experiments (submitted version)

The above analysis of the dataset shows that Stage 0 can detect a real problem in a real dataset. However, it would be necessary to test the method in more cases to verify whether the rank criterion

works the same way in other situations. It is also necessary to verify whether this criterion offers any advantage over a simpler rule based on searching for pairs of variables with a log-log correlation close to 1. To confirm this, we use simulated data for which we know the correct result in advance. In this way, we can verify whether Stage 0 correctly identifies the relationship between the variables. The designs and responses were generated using the model $\log C_p = \theta_0 + \theta_{\text{Re}} \log \text{Re} + \theta_{\text{Ma}} \log \text{Ma} + \theta_{N_g} \log N_g + \varepsilon$. In this model, we set $\theta_{N_g} = 0$. This means that we know gravity has no real effect on the response in the simulated data. The simulations use random facilities, but all remain within the same control variable space (Ω, R, p, T) used in Table 5. The complete code and simulation results are available in the supplementary material.

There are two important points we must clarify. First, the model used to generate the data has a single global intercept and does not use different prefactors for each geometry, as is the case with the actual data. This allows us to study the mathematical properties of the rank and of κ separately, without other effects related to the adjustment for geometry influencing the result. These effects are directly verified using the real data in Section 5.1, by analyzing the rank of the combined data after removing the effect of each geometry. Second, we consider that an estimated effect $\hat{\theta}_{\text{Ma}}$ (or the corresponding coefficient in Experiments C–D) has been correctly recovered when it satisfies $|\hat{\theta} - \theta| < |\theta|/2$. This means that the estimated value has a relative error of less than 50% with respect to the true effect introduced in the simulation. The main results do not vary significantly if a tighter threshold of 30% is used.

Attribution is the discovery algorithm’s prior, not the data’s. In a design similar to that of Liu2024, the three groups Re, Ma, N_g have rank 2. This means that the design does not have enough information to distinguish the three groups independently. To verify this, two possible structures are compared: Re, Ma and Re, N_g . Both structures yield exactly the same SSE and the same BIC, accurate to six decimal places; however, when the incorrect structure is used, the estimated value satisfies $\hat{\theta}_{N_g} = \theta_{\text{Ma}}/2$, as predicted by Proposition 1(ii), because $z_{N_g} = 2z_{\text{Ma}}$. Therefore, the quality of the fit does not allow us to determine which of the two structures is correct. Both explain the data in the same way. For this reason, it is necessary to use information from the experimental design. Stage 0 allows us to detect this limitation before fitting the model. If a new experimental condition is added using two different temperatures in the chamber, the relationship between the groups changes and the degeneracy disappears. With this new design, the groups can be distinguished, and the analysis correctly recovers the true structure.

Practical identification tracks κ , not just rank. In more than 400 full-rank, two-group randomized designs, the error in estimating θ_{Ma} increases as κ increases. This relationship is clear and statistically significant (Spearman’s correlation $\rho = 0.706$, $p = 1.2 \times 10^{-61}$). Two distinct situations can be identified here. When $\kappa < 30$, the median relative error is 17%, with an absolute error of 0.020. In contrast, when $\kappa > 30$, the absolute error increases to 0.336, which is approximately 2.8 times the true effect. These results show that a high value of κ produces less accurate estimates than low values of κ . Hence, the cutoff of $\kappa = 30$ used by Belsley et al. is not only a conventional rule but also proves useful in these simulated data.

With two candidate groups, the rank/ κ screen and a pairwise correlation screen are the same statistic. When comparing two centered and standardized variables, the relationship between the correlation r and the condition number κ is given by $\kappa = \sqrt{(1 + |r|)/(1 - |r|)}$. This relationship was confirmed numerically with an error of less than 7×10^{-7} . In particular, $|r| < 0.99 \Leftrightarrow \kappa < 14.1$. In a comparison involving 3000 tests, the rule based solely on correlation was slightly more accurate in this case: 0.883 versus 0.841. This indicates that, in designs with only two groups, the correlation-based measure performs slightly better. We present this result, even though it is negative, because in the case of two groups, we did not find an advantage of the criterion based on rank and κ over the simple correlation-based rule.

With three or more groups, pairwise correlation is blind to joint degeneracy. A group may depend on two others even if none of the pairwise correlations is greater than 0.99. This is exactly what happens in the case of Xia2024 described in Section 5.1: the three groups satisfy the relationship $z_{N_g} + z_{Re} - 3z_{Ma} = 0$, but no pairwise correlation reaches 1. Over 4000 randomized designs with three groups, this situation occurred in 23.2% of the cases. In these designs, the median number of joint conditions was $\kappa = 306$, even though all pairwise correlations were less than 0.99. However, only 15.8% of these cases could correctly recover the true structure. The rule based solely on pairwise correlations considered all these cases to be reliable, even though many of them were not. The criterion based on rank and κ was more accurate: it achieved an accuracy of 0.720, compared to 0.376 for the correlation-based rule. The false positive rate was 28% for the rank/ κ criterion, compared to 62% for the correlation rule. This rate indicates how often the method considers a design to be adequate when it is not. The ability to detect problematic cases was similar for both methods, with a recall of 0.534 for rank/ κ and 0.590 for the correlation rule. Both methods classify the design as NOT-IDENTIFIABLE with similar frequencies. The main difference lies in how reliable the cases are that each method considers identifiable.

9 S9: Complete verdict case analysis

This is the full case analysis of the verdict rules summarised in Section 3.7 of the main text.

Figure 2 clearly illustrates how the decisions in Algorithms 1–2 work. The decision tree combines the results of the four components and leads to one of the four final outcomes. Some steps of the algorithms 1–2 are presented in summary form because a single line of pseudocode does not allow all the necessary conditions to be explained. *Why NON-TRANSPORTABLE applies to only one object, and which object that is:* the adjusted law $\hat{g}$ (which includes the common exponent q , the prefactors for each geometry and the rule used for an unobserved geometry) and any structure S are evaluated separately. They are never combined into a single label.

NON-TRANSPORTABLE applies only to the fitted law $\hat{g}$ as a whole when its own R^2 in the LOSO test is ≤ 0 and the CONFOUNDING-SUSPECTED flag is set. Algorithm 1 calculates a single LOSO value for $\hat{g}$, so this label does not apply to a structure S . We intentionally consider $\hat{g}$ as a whole, and not just the ‘exponent’. The LOSO R^2 measures how well the complete model performs when attempting to predict for a facility that has been left out (Table 7). This result depends both on the common exponent and on the prefactors for each geometry, as well as on the rule used to predict a geometry that had not been seen before. When we say that $\hat{g}$ is NON-TRANSPORTABLE, we are saying that the complete law, as it has been fitted and is used to make predictions, does not perform well when applied to a new facility. We are not specifically saying that the exponent q is caused by the facility. Such a claim would be more specific than the LOSO results allow.

We do not regard this result as a FACILITY-ARTIFACT. According to Definition 1, for there to be confusion between facilities, two conditions must be met simultaneously: that the model behaves similarly to a model that uses only the facility, and that its performance deteriorates when a facility is omitted (LOSO). In the case study we present, only the second condition is met. The model $\hat{g}$ clearly performs better than a model that uses only the facility when evaluated with the fit data (§5.6). Therefore, Definition 1 does not apply in this case. Nor would it be correct to use the label FACILITY-ARTIFACT, because an $R^2 \leq 0$ on its own does not prove that the result is an artefact caused by a facility. We use the label NON-TRANSPORTABLE. This label best describes what the test shows: the fitted law does not perform well when used to predict at a facility that has been left out of the fit. This does not mean that the law is simply identifying the facility.

Structure S has its own results within the discriminant tests. It is considered SUPPORTED-

UNATTRIBUTED when (Tests 3–4 = PASS) $\wedge$ (Tests 1–2 = FAIL). If all four tests are passed, no negative label is assigned. In any other case, the result is INCONCLUSIVE. This also includes cases where $F = 1$, because in that situation Test 4 cannot be carried out. If the law $\hat{g}$ is not labelled NON-TRANSPORTABLE, its result is determined using logic based on the Ibragimov–Müller interval and the physical regime shown in Fig. 2. This part of the analysis does focus specifically on the exponent q , as it checks its Ibragimov–Müller interval and whether it belongs to the established physical regime. Part of the Track A logic analyses the exponent q , although the label NON-TRANSPORTABLE refers to the complete law $\hat{g}$. When there are several results for different objects and it is necessary to summarise them under a single label, we use the least favourable result rather than calculating an average.

Conformal cap: The CONFORMAL-COVERAGE-FAILS label is not a final result. It is used when any of the three conformal conditions set out in the protocol are not met: LOGO-min, n_{eff} and the maximum weight of a group (Table 4). It does not depend solely on LOGO-min, but on all three conditions. When any of these conditions is not met, a REPORT-AS-PHYSICS conclusion is downgraded to REPORT-WITH-DISCLAIMER; that is, the result may only be presented with a warning. If a negative result already exists, this flag serves only as supporting evidence and does not alter that result. To calculate these measures correctly, the data from the group that is excluded are not used to fit the model or to calculate the calibration quantile that is subsequently used to evaluate it. Section 5.8 explains this procedure and why we do not consider it a valid formal guarantee for any type of data. We use it as an empirical test to check whether the method works well when applied to data that were not used in the fitting. *Design-not-identifiable cap:* This is verified separately for each object, $\hat{g}$ and S , following the two independent branches in Fig. 2. If the physical interpretation of an object depends on a group marked as DESIGN-NOT-IDENTIFIABLE, the positive result for *that object* is restricted: REPORT-AS-PHYSICS changes to REPORT-WITH-DISCLAIMER. This does not change a NON-TRANSPORTABLE or SUPPORTED-UNATTRIBUTED result that has already been assigned to either of the two objects. A signal caused by one object does not limit the result of the other. *Rule for $F = 1$:* The purpose of Definition 1 and Component 3 is to verify whether a result holds across different facilities. When there is only one facility, the facility-level LOSO and CI^{IM} ($F-1 = 0$ degrees of freedom) cannot be calculated. It is not simply that they are uninformative: they are mathematically undefined because there is no second facility to compare them to. An earlier version of the algorithm calculated them anyway and changed the final label afterward. This is different from not calculating a quantity that is undefined. With $F = 1$, the current algorithm completely omits these two steps when they should be performed, and does so only for $\hat{g}$. The evaluation of S is an independent step. It is performed whenever S is selected, regardless of the value of F . It is not part of the branch for $F = 1$. An earlier version of the algorithm contained this error and only evaluated S when $F > 1$, even though the text stated otherwise. When $F = 1$, Test 1 uses the *geometries* from the dataset instead of the facilities as groups. This is an additional check and not a test of generalization across facilities. It is presented as an auxiliary diagnostic. Tests 2–3 are performed in the same way as in the other cases. Test 4 (LOFO) is omitted because it requires more than one facility. Therefore, when $F = 1$, the result for S is limited to INCONCLUSIVE. It cannot be labeled SUPPORTED-UNATTRIBUTED or receive a fully positive result, because both require that Test 4 has passed. The result of $\hat{g}$ is restricted to REPORT-WITH-DISCLAIMER (SINGLE-SOURCE ONLY) when $F = 1$. The fact that a lack of portability is not detected with a single facility does not prove that the result works on other facilities. Therefore, it should not be presented with the same label as a result obtained when $F > 1$. *S passes the four tests:* it yields no negative signals and is presented together with the fitted law $\hat{g}$, including its common exponent q , taking into account the two previous constraints. This protocol does not calculate Ibragimov–Müller-style confidence intervals for the parameters

specific to S (q_1, q_2, m separately). This is a limitation of the protocol and is left as future work in Section 6.4. NO STRUCTURED CANDIDATE and S : INCONCLUSIVE are internal states used by Track B to evaluate the structure S . They are not additional results. The four main results of the protocol are: REPORT-AS-PHYSICS, REPORT-WITH-DISCLAIMER, SUPPORTED-UNATTRIBUTED, and NON-TRANSPORTABLE. $\hat{g}$ can receive either REPORT-AS-PHYSICS or REPORT-WITH-DISCLAIMER. The label NON-TRANSPORTABLE applies only to $\hat{g}$, while SUPPORTED-UNATTRIBUTED applies only to S . If no S structure is selected, or if its result is INCONCLUSIVE, the final result of the analysis is still one of the four main results. The decision depends solely on Track A ($\hat{g}$). The DESIGN-NOT-IDENTIFIABLE signal from Stage 0 limits the physical interpretation from the outset, before the two branches of the analysis are run. It does not appear as a separate result in Figure 2.

The error each component prevents in the case study

Table 7: The error each component prevents in the case study of Section 5. This table is an illustration drawn from the same data that motivated the protocol; the systematic evidence that the combination is needed is the component ablation on simulated data with known ground truth (§5.9, Table 14).

Component removed	Error that would result	Evidence
Stage 0: Design-identifiability screen	A residual Ω -dependence could be attributed to compressibility (Ma) rather than centrifugal loading (N_g) without disclosing that the pooled, geometry-adjusted design is ill-conditioned for that contrast	§5.1: $\kappa = 88.7$ for (Ma, N_g) after geometry adjustment
C1: Honesty suite	A weak SR engine’s poor fit could be mistaken for domain confounding, rather than ruled out as the cause	§5.3, $R^2=0.979$ recovery
C3: Few-cluster-robust inference (Ibragimov–Müller)	The by-points interval $[-0.180, -0.033]$ would be reported as a statistically established exponent	§5.5, IM interval $[-0.362, +0.071]$ includes zero
C2: Discriminant battery	S_5 ’s nominal BIC-Laplace evidence ($\text{BF} = 5.8 \times 10^5$, 69.2% stability) would be read as a genuine compressibility transition	§5.7: Tests 1–2 show it is support-confounded, not established as physics
C4: Normalized conformal	The R^2 -based transfer failure (Table 7) would stand without a second, differently-constructed corroborating route; an implementation without per-fold cross-fitting reports 80% worst-group coverage for an unseen geometry instead of 0%	§5.8: honest LOGO-min collapses $0.800 \rightarrow 0.00$, an independent confirmation of Table 7’s negative R_{LOSO}^2

10 S10: Structure selection for S_5 , extended

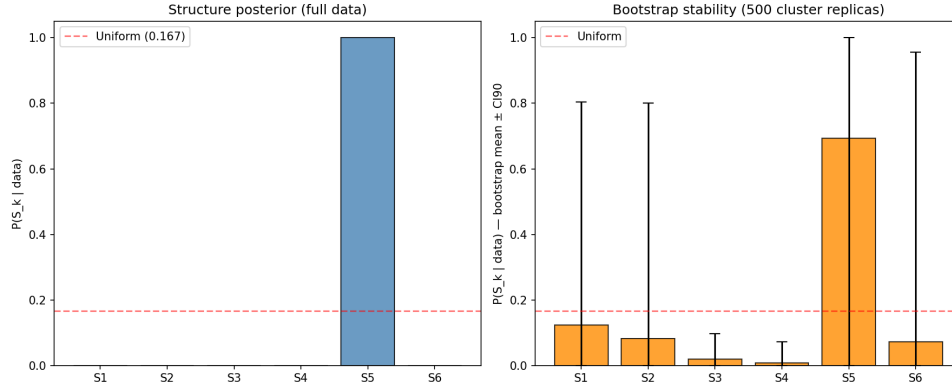


Figure 3: Bayesian structure posterior with bootstrap stability. S_5 is selected decisively and is stable in 69% of replicates — the profile of a structure that standard model-selection criteria would retain absent the audit.

Non-monotone stretch of the threshold sweep. Of the 32 points with $0.19 \leq \text{Ma} < 0.23$ that cross the threshold together between the sweep’s $m=0.19$ and $m=0.23$ rows, 23 (72%) are a single geometry (`Vrancik_alternator_shrouded`, repeated measurements at near-identical nominal Mach ≈ 0.21 – 0.22 , verified against the underlying data): the Bayes factor at $m=0.23$ reflects that one geometry’s block of repeated measurements entering the low-Mach regime as a unit, not a smooth trend in m . This is consistent with, not a caveat on, the reading above: a Bayes factor this sensitive to which discrete block of repeated measurements a threshold happens to fall on either side of is additional evidence against reading the sweep as tracing a physical transition.

11 S11: Specificity experiments of the submitted version

Caveat. These experiments are kept for completeness. While re-implementing the protocol for the revision we found that experiments (ii)–(iv) below did not run Algorithm 1: they used a simplified classifier (`run_framework_verdict` in `scripts/remediation_experiments.py`) that replaces the Test 1 rule by an asymptotic χ^2 test and omits Tests 3–4. Their false-positive and power figures therefore describe that classifier, not the protocol. They are superseded by the simulation benchmark of Section 5.9 of the main text, which runs the protocol as specified there (model selection between S_1 and S_5 only; no Stage 0; no single-facility branch). The Bayes-factor figures of (iv) (a value as large as the observed one in 1 of 200 null replicates) do not depend on the classifier and are the ones quoted in Section 5.7.

To verify that the detector does not always flag a result as an “artifact,” a test was conducted using data in which there is no mixing between different facilities. *(i) Real data from a single facility.* In this case, only the Vrancik1968 dataset was used, and the internal geometry was taken as the clustering criterion. Since there is only one facility ($F = 1$), the facility-LOSO and CI^{IM} are not calculated, as these methods are not defined for this case (rule $F = 1$ of Algorithm 1). The SRSD-Feynman recovery test can be performed because it does not depend on F and passes the test. Furthermore, the structure search does not associate the facility with any regime. More than one facility is needed to verify whether a law also holds in other facilities. For this reason, it is not possible to evaluate invariance across facilities, which is one of the properties intended to

be verified via Definition 1 and Component 3’s few-group inference (rule $F = 1$ of Algorithm 1, §3.7). Therefore, the result is presented as REPORT-WITH-DISCLAIMER (SINGLE-SOURCE ONLY) and not as REPORT-AS-PHYSICS. The protocol also does not consider $\hat{g}$ to be NON-TRANSPORTABLE, because with a single facility, there are no other facilities with which it can be compared. This result allows us to confirm that the detector does not generate a non-transportability signal when such a comparison is not possible. Although this problem is not detected when $F = 1$, this does not prove that the law can be generalized to other facilities. To prove this, data from more than one facility would be necessary. The result’s label indicates this limitation. (ii) *Simulated data with no facility-related confounding.* $K = 200$ experiments were also conducted using simulated data. In all of them, the same real-world distribution was used, and only the distribution of the covariates was changed, without introducing any facility-related effects. These simulated data were used solely to determine how the simplified classifier performs under these conditions. The false positive rate during the 200 experiments was $\widehat{\text{FPR}} = 0.010$, with a 95% confidence interval of $[0.001, 0.036]$. This experiment was presented in the submitted version as evidence that the detector can distinguish between a real result and a possible “artifact,” since the value does not exceed 0.05. This result must be interpreted within its limitations. The test only shows how the simplified classifier performs under the $K = 200$ simulated conditions used here, all of which are based on the same real-world law and on a change of covariates with no effect associated with the facility. The data were generated specifically for this study; therefore, it cannot be claimed that the detector has this same specificity for any type of data. (iii) *Power and its limitation.* The detector has low ability to identify the problem when the confounding is introduced through a change in the *prefactor* C_i for each facility. Across the $K = 200$ experiments, power remains between 0.065 and 0.085 for $\Delta \in [0.1, 0.5]$, with approximate 95% Clopper–Pearson confidence intervals of $[0.04, 0.13]$ for each value (Fig. 4). The situation changes when the problem is in the structure itself. In this case, each facility has a different exponent and, in addition, its Mach range is separated from those of the other facilities. The detector performs better under these conditions: power increases from 0.13 when there is no difference between the exponents to 0.70 when the difference reaches 0.20. This type of confounding is commonly studied in the literature on domain generalization. This case does not correspond to the mechanism found in S_5 in Section 5.7. In S_5 , the Mach threshold does not clearly separate any particular facility from the others (Supplement S6). The value of 0.13 when there is no difference between the exponents should also be taken into account. It cannot simply be treated as a reference value because it represents a *second false-positive rate*. This value is approximately ten times higher than the 1% obtained with the clean data. The reason is that, even without introducing a difference between the exponents, the simulated data still contain a separate Mach range for a single facility. The structure of the data affects the specificity of the simplified classifier. When the covariates normally overlap across facilities, the false-positive rate is approximately 1%, but when a single facility has a separate covariate range from the others, this rate increases to approximately 13%. This second scenario is more similar to the support problem that appears in the S_5 case than the test performed with clean data. This limitation means that the simplified classifier can detect confounding when one facility has a covariate range that is clearly different from the rest. However, it has low ability to detect a change that only affects the *prefactor*, because the model used for each group already accounts for this type of change. These results should not be interpreted as a specific validation of the protocol for the S_5 case. In this case, the problem is identified through the contingency and threshold-change tests included in the discriminant tests (Section 5.7). The problem is related to an imbalance in the data support: there are 8 points compared with 106 points in the combined dataset. Neither this power test nor the Stage 0 rank/conditioning diagnostic is designed to directly detect this type of imbalance. The result related to the *prefactor* is clearly shown because it helps explain a limitation of the method. In class-SR, each geometry has its

own *prefactor*, so any change related to this value is *absorbed by the model itself*. Therefore, if the issue affects only this value, the model itself can adapt to it, and the audit does not detect it as a change in the structure. This is a limitation of the method, since the protocol cannot detect every type of confusion. Supplement S6 analyzes another case in which the confusion affects the structure and not just the *prefactor*. Here, the Reynolds number changes for a single facility, and this facility has a different Mach number range than the others. This represents the typical case in which a single facility behaves differently. It differs from the combined support problem observed in Section 5.7. In this case, the study examines how the detector’s performance changes as the difference between the Reynolds numbers increases. (iv) *Null calibration under the S_5 case’s actual mechanism*. Neither (ii) nor (iii) mimics the type of imbalance that appears in the S_5 case. In those tests, ranges of covariates are either separated across facilities or distributed in a more orderly fashion. In contrast, the real data contain a small portion of the data points—8/106—that correspond to low Mach numbers and are scattered (Table 11). To study this situation directly, we used the 114 real data points and retained their Reynolds numbers, Mach numbers, geometries, and facility labels. In this way, we ensured that all simulations exactly preserve the same 8/106 support imbalance found in the original data. From these data, we generated new y values under the null model S_1 , which assumes a single common exponent. To generate the data, we used the adjusted real-world value $q = -0.0745$, the prefactors for each geometry, $\hat{\sigma}_g$, and i.i.d. Gaussian noise. We then applied the same threshold search and Bayes factor calculation procedure used to find S_5 (`analysis_identifiability/s5_null_calibration_real_support.py`) to the $K = 200$ replicates. For this test, we used an independent, simplified classifier designed to roughly follow the logic of Track B. Therefore, the procedure does not directly execute Algorithm 1 on S . With this classifier, in 13 of the 200 replicates (6.5%), the null model is unable to correctly rule out the spurious regime. In these cases, the result was neither “no regime detected” nor an explicit rejection. Furthermore, a Bayes factor as large as that observed in the real data, $\text{BF}_{S_5/S_1} = 5.8 \times 10^5$, appeared in only 1 of the 200 replicates (0.5%). No replica reached the maximum value of 6.2×10^6 obtained in the search. The value of 6.5% merely demonstrates how the Track B classifier performs under this null model, which maintains the same support imbalance as the real data. This result does not change the $\text{FPR} = 1\%$ obtained with the simplified classifier and the clean data in (ii), because they are different tests and use different null models. This calibration complements the permutation test in Section 5.7. Both tests use the same observed data and the same test statistic. The difference is that this calibration preserves the same support imbalance found in the actual data, whereas the permutation test uses a data shuffling procedure. Both tests lead us to a similar conclusion: even when the same support imbalance as in the actual data is preserved, it is rare to obtain by chance an S_5 signal as large as the one observed. This test alone cannot determine whether the result corresponds to a real physical transition or to an artifact. The fact that the result is uncommon under the null model does not mean that one of the two possibilities is ruled out. A real physical transition could exist, but it could also be an artifact not caused by chance and related to the way the covariates are distributed. In this test, the null model uses a single exponent and does not include regime changes, so it cannot distinguish between these two possibilities. To make this decision, Discriminant Tests 1–2 from Section 5.7 are used. These tests analyze whether the obtained threshold is independent of the facility and whether it coincides with the expected physical onset. Therefore, the conclusion is not based solely on how rare it is to obtain this signal under the null model. Later on, a specific check of the data quality could be added to Stage 0. This would make it possible to detect from the outset whether there is an imbalance that could affect the identification of the effect. Thus, problems such as the one observed could be identified before model selection takes place, rather than afterward through the discriminant tests. The results of (ii)–(iv) describe the simplified classifier, not the protocol: with the cleaned data its false positive

rate was low, $\text{FPR} = 1\%$. They are historical; the operating characteristics of the complete protocol are those of Section 5.9 of the main text. The results indicate that the signal found in S_5 does not appear to be due simply to chance. However, this test does not allow us to distinguish whether it is a real physical phenomenon or an artifact. To do so, we must use the tests mentioned earlier.

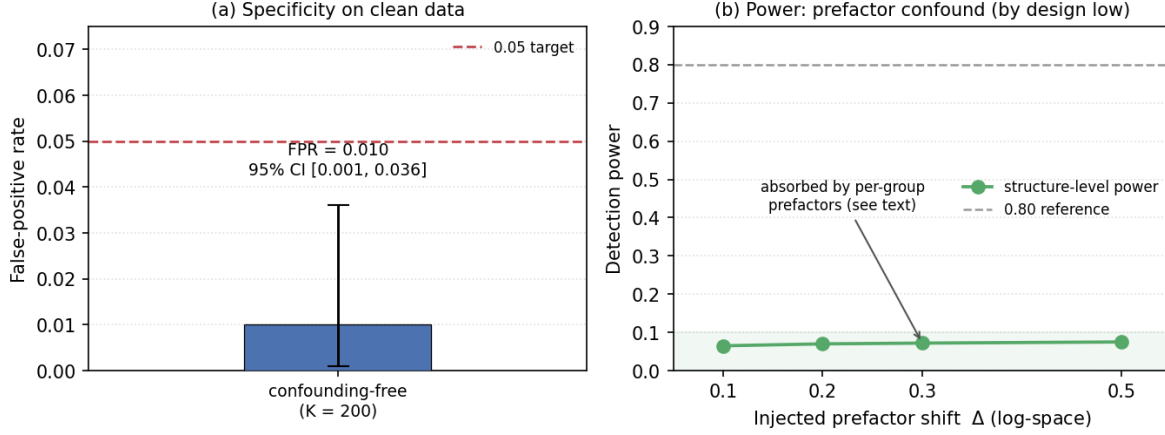


Figure 4: Operating characteristics of the simplified classifier of the submitted version (superseded; see the caveat above). (a) False-positive rate on confounding-free simulated data ($K = 200$) with 95% CI; the simplified classifier fires on 1% of clean datasets, well under the 0.05 target. (b) Detection power for a prefactor-level confound is low by design, because class SR absorbs per-group prefactors; the audit targets structure-level confounding (the S_5 case), a scope we state explicitly.

12 S12: Notes on reproducibility: the conformal leakage and its correction

This section documents, for transparency, how the held-out-geometry conformal evaluation was first computed with a leak, how the leak was found, and how it was corrected. The final numbers are in Table 12 of the main text.

The quantification of uncertainty is described in Tables 8 and 12, and in Fig. 5. First, the original calculation of this component is presented, followed by the filtering of information identified by the external review. The corrected result changes the conclusion.

What was originally reported. The acceptance criterion established above ($\text{LOGO}_{\min} \geq 0.80$) was evaluated by excluding one geometry at a time ($G = 12$). The split conformal prediction, which assumes that the data are interchangeable, performed very poorly under this change: the minimum LOGO coverage was 0.167 compared to the nominal value of 0.90, due mainly to the fold corresponding to the smallest geometry (`Vrancik_cylindrical_in_stator`, $n=6$). Mondrian and the normalized conformal methods were reported as meeting the criterion with an exact value of 0.800.

The leak. The two results that appeared to meet the criterion were actually caused by information leakage, as confirmed by reviewing the code used (`sr_engine/conformal.py`, `scripts/run_conformal_prediction.py`). This issue was then corrected with a new cross-fitting script. There were two different problems:

1. The “LOGO-min” presented for Mondrian does not actually involve leaving out an entire geometry, and is more limited than its name, “leave-one-point-out,” might suggest. To obtain this calculation, each geometry is taken, and its minimum coverage is calculated after removing the nonconformity score of that specific point before calculating the calibration quantile. However, that point and the other points of the same geometry remain part of the fit that generated $\hat{g}$, C_i , and $\hat{\sigma}_i$. Only the point is left out when calculating the quantile. Furthermore, no complete geometry is left out of the fit. This procedure does not allow us to assert that coverage exists for an *unobserved* geometry, nor should it be interpreted as a stronger guarantee within the same domain. In Table 8, it is renamed and is no longer compared to the LOGO acceptance threshold.
2. In the LOGO evaluation of normalized conformal fitting, the points from the test geometry were excluded from the calibration set but were evaluated using a predictor and a scale, $\hat{g}$ and $\hat{\sigma}_g$, that had been fitted to the 114 points of the complete set, including the test geometry itself. This is important because the prefactor C_i and $\hat{\sigma}_i$ for each geometry in class-SR are free parameters that are estimated based on the mean and variance of the residuals for that same geometry. The situation is similar to a substitution where the points from the geometry that was supposedly excluded were evaluated using a model that had already utilized information from that geometry.

The correction. The LOGO at the geometry level was again calculated using strict cross-fitting for each fold. For each excluded geometry, the common exponent and the geometry-specific prefactors are re-estimated using only the other 11 geometries (`class_sr_fit` is refitted with $\mathcal{D} \setminus \mathcal{D}_g$). The prediction for the excluded geometry uses the same average prefactor across geometries and the same average $\hat{\sigma}$ that had been established as a fallback for unobserved geometries in Table 7. The coverage decreases more than usual: the LOGO minimum drops from the originally reported 0.800 to 0.00 (`CHIEF_chamfer_arm`, $n=5$). The coverage-weighted average for the 12 excluded geometries is 0.009: 11 of the 12 geometries have no test points within their interval, except for the twelfth (`ZJU400_enclosed_arm`), which includes 1 of 8 points. The reason for this result is the same as that already shown in Table 7. Using the average prefactor for all geometries does not allow for a good prediction, since the prefactors of the different geometries differ by a factor of approximately 70 in the original units ($\log C_i$ ranging from -5.00 to -0.68). Therefore, when attempting to predict a geometry that has not been observed, the residual relative to this mean value typically falls between 5 and 19 standard deviations within the geometry relative to zero. These differences prevent the result from falling within an interval obtained through calibration with a nominal level of 90%. The conformal variants evaluated do not meet the established threshold of $\text{LOGO}_{\min} \geq 0.80$ when dealing with a truly novel geometry. This result, with a more limited finding, helps us make a positive or negative decision.

The outer fold alone is not a complete conformal guarantee; we add the missing inner split. An independent review pointed out another issue to us: leaving the test geometry out of the fit is a necessary condition, but on its own it is not technically sufficient to guarantee distribution-free coverage. This limitation is supported by the conformal prediction literature [4, 6]. Across the 11 geometries used for training, the quantile used for calibration was still obtained from the same model with which the resubstitution residuals had been computed in the inner stage, even though the split performed at the outer stage was adequate. The procedure was repeated with a full nested split using `analysis_identifiability/conformal_logo_nested_crossfit.py`. First, the external split is maintained, leaving the test geometry out of the fitting. With the remaining

11 geometries, 200 random splits were performed, dividing the data into two equal parts: one for training and the other for calibration. In each run, `class_sr_fit` was trained using only the training set, and the other set was used to obtain scores without having been included in the training. All the scores from the 200 repetitions were combined to obtain the calibration quantile. The procedure is now truly out-of-fold at both levels, since no point is ever evaluated using a model that was trained on that same point.

What this construction does and does not establish. The external review correctly pointed out that a distinction must be made between being ‘out-of-fold’ and being valid according to conformal theory. This nested division does not correspond to the split conformal method or to CV+/jackknife+, where the excluded point is evaluated using the *same* fold-specific model that produced its calibration residual. These constructs are defined in [1, 4, 6]. The scores are not independent because the aggregated calibration quantile combines scores from 200 different fittings during internal training. During training, two things can happen: each point may appear in many internal calibration halves across the various divisions, and points that fall within the same division share the corresponding fitting. The excluded geometry, unlike CV+/jackknife+, is evaluated using another single final fit, distinct from all the models that produced a calibration score. We present the values 0.00 and 0.202 as an empirical (out-of-fold) *stress test* to evaluate whether the result can be transported to new geometries, and not as a certified conformal interval. For this reason, we do not claim that this procedure provides a formal, distribution-free $1 - \alpha$ coverage guarantee. What we can claim is that the procedure is honest in a narrower but useful sense: no point used to compute a calibration or test score has been evaluated by a fit that had previously seen that same point. If a formal conformal guarantee is needed, this result can serve as motivation for implementing a specific method, such as split conformal by geometry or group-aware CV+/jackknife+. This implementation is left as future work (§6.4), since we do not want to make a claim we cannot yet substantiate.

The new results do not change the conclusion obtained previously. Although the coverage estimate rises to 0.202 when the 12 folds are considered, the LOGO minimum remains 0.00 for the same geometry. This value of 0.202 is higher than the previous estimate of 0.009, but it is still well below the nominal target of 0.90 and the acceptance threshold of 0.80. The difference from the previous value is because we now know that 0.009 underestimated the true width of the calibration interval. The observed problem also does not change. When an observed geometry is not available, the mean computed across the different geometries as a fallback can deviate considerably from the true level of the new geometry. In this case, the difference reaches tens of standard deviations within each geometry, which explains why coverage remains so low. The new nested split allows the CONFORMAL-COVERAGE-FAILS flag in Algorithm 2 (§3.7) to be used as an additional confirmation, properly performed out-of-fold, of the lack of transportability. This conclusion had already been observed through the negative R_{LOSO}^2 value reported in Table 7. Therefore, this result should be understood as evidence supporting the audit’s central claim, rather than as proof that the evaluated procedure has a validated conformal capability. We also report both results, the one obtained with the outer split and the one obtained with the nested split, rather than showing only the final result. The difference between 0.009 and 0.202 is important because it shows the extent to which the earlier, outer-split-only computation understated the calibration quantile and therefore the coverage. We checked that this correction does not benefit normalized conformal alone. To do so, we repeated exactly the same nested procedure with split conformal (`analysis_identifiability/conformal_logo_nested_crossfit_split.py`). The LOGO-minimum was also 0.00, while its honest weighted-mean coverage was 0.053, even lower than that

obtained with the normalized method. The fix does not give any particular advantage to the method selected in our analysis.

Geometry-conditioned calibration does not transfer to an unseen facility either. For a calibration set (Section 5.8 shows that it is precisely for this reason that coverage decreases), a geometry excluded during a test is entirely new to the calibration set. An excluded geometry is still represented in the calibration by the *facility* to which it belongs, via *other* geometries from that same facility. The excluded facility itself has neither of these: none of its geometries, and therefore nothing from the facility itself, appears in the calibration. Another case was studied in which a *facility* was isolated to evaluate the split conformal model. The result is that the minimum LOSO coverage falls to 0.439 (Vrancik1968), calculated from the same globally fitted residuals (not via cross-fitting) as the original LOGO value, now corrected. This is not interpreted as an exact value, and it is not used as evidence: it carries the same leak as the original LOGO value and was not recomputed with cross-fitting, so no bound can be inferred from it. The (adaptive) ablation of learnt clusters from Table 8, evaluated by its own construction at the facility level, yields the same minimum value of 0.439; since that evaluation has the same leak, it is not independent corroboration either.

Mondrian and normalised conformal calibration are geometry-dependent, and the geometries of an excluded facility do not have their own calibration points; therefore, they cannot be evaluated whilst leaving out an entire facility without introducing an ad hoc alternative rule. The cross-fitted held-out-geometry evaluation shows that geometry-conditional normalisation does not meet the acceptance criterion for a new geometry; a new facility brings new geometries as well, but we have no honestly refitted facility-level evaluation, so we do not draw a conclusion for that case.

Marginal coverage remains informative, but is a different, narrower claim. Marginal coverage never excludes a group from the fit; for this reason, it is not influenced. This is because the points already reflected in the calibration set fall within its interval on average, without implying that this result can be generalised to an unobserved group. To refer to this as ‘valid coverage’ or a ‘calibrated interval’ would be a mistake, as these terms could be interpreted as offering a guarantee for an excluded group. Within a fit, Mondrian (0.850) and the normalised method (0.911) are still considered informative diagnostics. The effective sample size of the normalised conformal method ($n_{\text{eff}} = 75.2$ out of 114, with no calibration point exceeding 3.25% of the total mass of weights) remains a valid and unbiased diagnostic of the concentration of its calibration weights. It is no longer necessary to present this alongside a satisfactory LOGO-min; we simply retain the marginal coverage in Table 8. The comparison could lead to a misinterpretation of the results, as it could previously have been interpreted as a stronger assertion than each result individually supports. The diagnosis is carried out within a fit, calculated on the basis of the coverage counts for each point rather than from a separate group. Table 10 presents exact (Clopper–Pearson) 95% confidence intervals for the marginal coverage per facility. The two smallest facilities (Liu2024, $n = 20$; Zheng2024, $n = 8$) stand out from the rest and show that the width of these intervals may conceal considerable uncertainty even for this more limited claim.

Table 8: Conformal prediction under inter-facility shift, geometry partition ($G = 12$): original computation and its known leak. Marginal coverage (col. 2) is a within-fit diagnostic, unaffected by the leak described in §5.8. LOGO-min (col. 4) is the held-out-geometry diagnostic *as originally reported*: it reused a globally-fit predictor and $\hat{\sigma}$ across all folds (external review identified this as leakage, confirmed against the code) and must *not* be read as a held-out result — the honest, leak-corrected re-evaluation is in Table 12. Mondrian’s figure is relabeled: it is a within-domain leave-one-*point*-out minimum (the point stays in the fit of $\hat{g}$, C_i , $\hat{\sigma}_i$; only its own score is left out), not a held-out-*geometry* figure, and is not comparable to the other rows’ LOGO-min even before correction. The learned-group ablation was not re-run with the nested correction (§5.8, §5.7(iv)); its LOGO-min stands only as originally computed.

Variant	Cov.	Width	LOGO-min (orig., leaky)	n_{eff}
Split (vanilla)	0.918	0.714	0.167	114.0
Mondrian (geom, within-domain LOO [†])	0.850	0.472	0.800 [‡]	114.0
Normalized	0.911	0.473	0.800	75.2
Learned-group (abl., not re-run [‡])	0.914	0.590	0.439	114.0

[†]Not a leave-one-geometry-out figure; see the table caption above. [‡]Not covered by the nested cross-fit correction of Table 12; this LOGO-min carries the same leak as this table’s other rows and should not be compared against Table 12’s figures.

Table 9: Conformal prediction under inter-facility shift, geometry partition ($G = 12$): honest, leak-corrected re-evaluation (§5.8), for the two variants we re-ran under the nested construction (outer geometry-fold, inner out-of-fold calibration split, re-estimating the exponent, every prefactor, and every $\hat{\sigma}$ per fold, excluding the held-out geometry entirely). We re-ran both normalized and split (vanilla) conformal, so the correction is not confined to the variant it happens to favor: both collapse well under the 0.80 bar (LOGO-min 0.00 for each), and split’s honest weighted-mean coverage (0.053) is markedly *worse* than normalized’s (0.202), the opposite of what a reader might expect if the correction were somehow flattering to the paper’s preferred variant. Mondrian and the learned-group ablation are not included here: neither was re-run under this correction (see Table 8).

Variant	LOGO-min (honest, nested cross-fit)	Weighted-mean cov. (honest, nested)	n_{eff}
Split (vanilla)	0.00	0.053	114.0
Normalized	0.00	0.202	75.2

Table 10: Marginal per-facility coverage under the normalized-conformal variant, with Clopper–Pearson 95% binomial confidence intervals computed from the underlying per-point coverage counts (data: Section 5.8); these treat per-point coverage indicators as independent Bernoulli trials and should be read as descriptive, not as inference that accounts for the shared fit, shared calibration set, and within-facility dependence. Like the marginal-coverage column of Table 8, this is a within-fit diagnostic and makes no held-out-group claim: it is not comparable to, and should not be confused with, the honest (cross-fit) held-out-*geometry* LOGO-minimum of Table 8 (0.00), which answers a different question (coverage for a group excluded from the fit) and collapses where this table’s numbers do not.

Facility	n	Coverage	95% CI (Clopper–Pearson)
Xia2024	45	0.911	[0.788, 0.975]
Vrancik1968	41	0.927	[0.801, 0.985]
Liu2024	20	0.900	[0.683, 0.988]
Zheng2024	8	0.875	[0.473, 0.997]

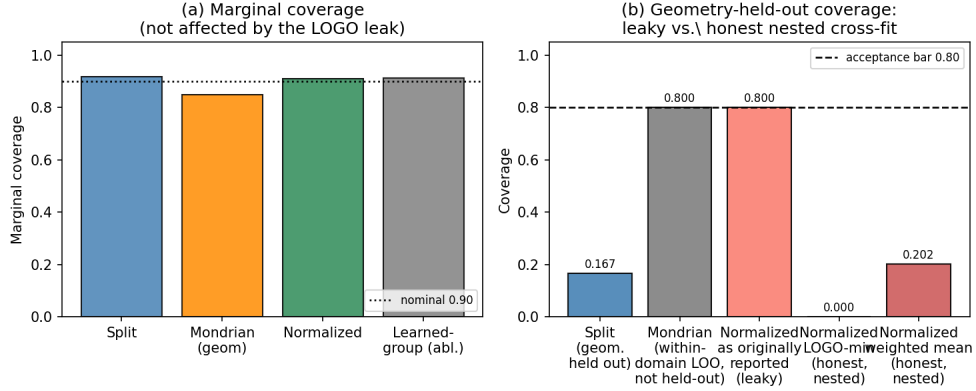


Figure 5: (a) Marginal coverage, a within-fit diagnostic unaffected by the LOGO leak described in §5.8. (b) Geometry-held-out coverage: split conformal’s worst-case coverage is 0.167; Mondrian’s originally reported figure is a within-domain leave-one-point-out minimum, not a held-out-geometry one; normalized conformal’s LOGO-minimum, originally reported as 0.800 (leaky: globally-fit predictor and $\hat{\sigma}$ reused for the held-out geometry), is 0.00 under honest, fully nested (outer and inner) cross-fitting — distinct from, and not to be confused with, the nested weighted-mean coverage across all 12 held-out geometries (0.202, the rightmost bar), which is higher than the minimum but still far under the 0.80 bar. Normalized conformal does not achieve the ≥ 0.80 acceptance bar (fixed in the final protocol) for a genuinely unseen geometry by either measure (§5.8).

Model-plus-fallback comparisons (full discussion)

These are model-plus-fallback comparisons, not capability-only comparisons, and we report them as such. None of the class-SR, IRM, or GroupDRO models can, on their own, make a prediction for a geometry they have not seen. To do so, each model requires an additional rule specifying which intercept to use for that new geometry, since it cannot estimate it from data that does not exist. In class-SR, the average prefactor of the other geometries is used as an alternative, as explained in the note for Table 7. This value is calculated from the estimated prefactors for the other geometries. IRM and GroupDRO use an intercept equal to zero, because

these methods do not produce an equivalent average value that can be used for a new geometry. These two rules are not the same. LOSO scores do not allow for a comparison of which model generalizes best on its own. Each model is compared together with the additional rule used to make predictions on new geometries. We consider the order of the results to be informative for one important reason: the comparison is not set up to favour class-SR. The rule of using a zero intercept is generally stricter than that of using an average prefactor. Under this rule, the estimated levels for the other geometries are completely ignored. Even so, class-SR performs much worse (-1.001 vs. $0.198/0.190$). If class-SR also used a zero intercept, its LOSO score would likely be even worse; in that sense the comparison is, if anything, favourable to class-SR. These results do not allow us to say exactly how much better or worse one model is than another. Class-SR performs better than classical physical correlations when attempting to make predictions for new facilities. Therefore, the problem is not that SR is worse than classical windage theories. This is the most important finding of this study: *no combination of model and prediction rule generalizes well to new facilities, and the more flexible models perform worse than the simplest model*. The fact that even cross-domain generalization methods designed specifically for this problem yield scores below 0.2 suggests that there are significant differences between facilities that cannot be resolved simply by changing the model. Due to the differences between the models and the rules used to make predictions about new geometries, as explained in section 5.6, this result should be considered only as evidence supporting this conclusion. A natural question is why not simply add facility as a fixed effect rather than run the full audit: Table 7’s “ C_p const. per facility” row *is* exactly that model (a facility-level intercept, $n_p = 4$), and it is not a fix — $R_{\text{orig}}^2 = 0.538$ and LOSO -1.199 , worse than the simple global power law with no facility term at all. A fixed-effects model can only capture differences in the level of the prefactor. As shown in Section S11, the audit has little ability to detect this type of difference. This approach cannot detect a *structural* problem caused by the data set, such as the one shown in S_5 . The problem lies not in the intercept, but in which regressors and which form of the function are selected to construct the model.

13 S13: Simulation benchmark (version 2), full tables

Generator: `revision/benchmark_v2.py`; implementation: `revision/audit_v2.py`; corrections to the first (frozen) implementation and expected outcomes recorded before the run: `revision/CORRECTIONS_v2.md`; per-run outputs: `revision/bench_v2_K20.jsonl`; analysis: `revision/analyze_v2.py`; these tables: `revision/gen_S13.py`. The first implementation (`audit_frozen.py`, `FROZEN_PROTOCOL.md`, $K=30$, five regimes) is kept unchanged and is not reported. Nine regimes $\times$ 16 cells $\times$ 20 replicates = 2,880 data sets; master seed 20261002. Effect levels (low/high): R2 facility-offset SD 0.15/0.40; R3, R3m, R3s exponent jump 0.05/0.15; R3w 0.005/0.015; R4a exponent shift of the banded facility 0.05/0.15; R4b, R4c additive artefact 0.15/0.40. Geometry prefactors $\log C_g = -2 + \mathcal{N}(0, 0.15^2)$; Mach $U(0.05, 0.35)$ except the R4a banded facility, $U(0.40, 0.65)$, and R4b, where all points are $U(0.12, 0.35)$ except the artefact subset, $U(0.05, 0.10)$; R3s uses the same Mach distribution as R4b (about 8% of the points of every facility moved to $U(0.05, 0.10)$, below the genuine threshold 0.11). Mach is drawn from distributions that span both sides of the threshold in every facility, but a given corpus need not contain points on both sides in every facility. Proportions are rounded half up to three decimals. Four regimes (R3w, R3s, R3m, R4c) and the corrections were added after the results of the first run had been seen.

Table 11: Track B (discriminant endpoint): fraction labelled STRUCTURE-SUPPORTED (P) and fraction with the expected label (E), by factor level. Expected: no STRUCTURE-SUPPORTED for R1, R2, R4a, R4c; SUPPORTED-UNATTRIBUTED for R4b; STRUCTURE-SUPPORTED for R3, R3w, R3m, R3s (R3s is anticipated to fail by design).

Regime	effect low		effect high		$\sigma = 0.03$		$\sigma = 0.08$		$n \times 1$		$n \times 2$		$F = 4$		$F = 6$	
	P	E	P	E	P	E	P	E	P	E	P	E	P	E	P	E
R1	0.025	0.975	0.013	0.988	0.025	0.975	0.013	0.988	0.019	0.981	0.019	0.981	0.025	0.975	0.013	0.988
R2	0.000	1.000	0.013	0.988	0.006	0.994	0.006	0.994	0.013	0.988	0.000	1.000	0.006	0.994	0.006	0.994
R4a	0.000	1.000	0.000	1.000	0.000	1.000	0.000	1.000	0.000	1.000	0.000	1.000	0.000	1.000	0.000	1.000
R4b	0.006	0.469	0.000	0.706	0.000	0.663	0.006	0.513	0.006	0.688	0.000	0.488	0.006	0.500	0.000	0.675
R4c	0.050	0.950	0.038	0.963	0.025	0.975	0.063	0.938	0.031	0.969	0.056	0.944	0.056	0.944	0.031	0.969
R3	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
R3w	0.969	0.969	1.000	1.000	1.000	1.000	0.969	0.969	0.969	0.969	1.000	1.000	0.981	0.981	0.988	0.988
R3m	1.000	1.000	0.994	0.994	0.994	0.994	1.000	1.000	0.994	0.994	1.000	1.000	0.994	0.994	1.000	1.000
R3s	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Table 12: Sensitivity to the Test 1 thresholds (single-facility share / minority share). Fraction labelled STRUCTURE-SUPPORTED (discriminant endpoint) in each regime, and fraction SUPPORTED-UNATTRIBUTED in R4a and R4b. The rule used elsewhere is 0.90/0.10. Changing the limit changes which datasets are SUPPORTED-UNATTRIBUTED, so verdict distributions differ slightly between 0.10 and 0.15 (R4b: see last column).

Thresholds	R1	R2	R4a	R4b	R4c	R3	R3w	R3m	R3s	S-U R4a	S-U R4b
0.80/0.05	0.019	0.006	0.000	0.378	0.044	1.000	0.984	0.997	1.000	0.031	0.003
0.80/0.10	0.019	0.006	0.000	0.003	0.044	1.000	0.984	0.997	0.000	0.031	0.588
0.80/0.15	0.013	0.006	0.000	0.000	0.041	1.000	0.984	0.997	0.000	0.031	0.591
0.85/0.05	0.019	0.006	0.000	0.378	0.044	1.000	0.984	0.997	1.000	0.031	0.003
0.85/0.10	0.019	0.006	0.000	0.003	0.044	1.000	0.984	0.997	0.000	0.031	0.588
0.85/0.15	0.013	0.006	0.000	0.000	0.041	1.000	0.984	0.997	0.000	0.031	0.591
0.90/0.05	0.019	0.006	0.000	0.378	0.044	1.000	0.984	0.997	1.000	0.031	0.003
0.90/0.10	0.019	0.006	0.000	0.003	0.044	1.000	0.984	0.997	0.000	0.031	0.588
0.90/0.15	0.013	0.006	0.000	0.000	0.041	1.000	0.984	0.997	0.000	0.031	0.591
0.95/0.05	0.019	0.006	0.000	0.378	0.044	1.000	0.984	0.997	1.000	0.031	0.003
0.95/0.10	0.019	0.006	0.000	0.003	0.044	1.000	0.984	0.997	0.000	0.031	0.588
0.95/0.15	0.013	0.006	0.000	0.000	0.041	1.000	0.984	0.997	0.000	0.031	0.591

Table 13: Track A verdicts by regime. Only R1 (transportable) and R2 (not transportable) have a truth defined in advance; the other regimes are descriptive.

Regime	NON-TRANSPORTABLE	REPORT-WITH-DISCLAIMER	REPORT-AS-PHYSICS
R1		0.047	0.953
R2		0.522	0.478
R4a		0.413	0.588
R4b		0.025	0.872
R4c		0.019	0.966
R3		0.659	0.331
R3w		0.031	0.903
R3m		0.578	0.416
R3s		0.453	0.500

Table 14: Track A component ablation: flag rate in R1 (false positives) and in R2 (detection of a pure facility shift).

Rule	R1	R2
(a) LOSO only	0.047	0.522
(b) IM only (CI includes 0 or not exactly one canonical regime)	0.109	0.081
(c) conformal only	0.994	0.994
(d) full Track A (not report-as-physics)	1.000	1.000

References

- [1] Rina Foygel Barber, Emmanuel J. Candès, Aaditya Ramdas, and Ryan J. Tibshirani. Conformal prediction beyond exchangeability. *The Annals of Statistics*, 51(2):816–845, 2023.
- [2] A. Colin Cameron, Jonah B. Gelbach, and Douglas L. Miller. Bootstrap-based improvements for inference with clustered errors. *The Review of Economics and Statistics*, 90(3):414–427, 2008.
- [3] Ivan A. Canay, Andres Santos, and Azeem M. Shaikh. The wild bootstrap with a small number of large clusters. *The Review of Economics and Statistics*, 103(2):346–363, 2021.
- [4] Jing Lei, Max G’Sell, Alessandro Rinaldo, Ryan J. Tibshirani, and Larry Wasserman. Distribution-free predictive inference for regression. *Journal of the American Statistical Association*, 113(523):1094–1111, 2018.
- [5] James G. MacKinnon and Matthew D. Webb. Wild bootstrap inference for wildly different cluster sizes. *Journal of Applied Econometrics*, 32(2):233–254, 2017.
- [6] Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random World*. Springer, 2005.